Attention Is All You Need

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TL;DR

Before 2017, translation models read a sentence **one word at a time** (RNNs), which is slow and makes far-apart words hard to connect. The Transformer throws out recurrence and uses **only attention**: every word looks at every other word at once. It set new records on English→German (28.4 BLEU) and English→French (41.8 BLEU) while costing less to train than earlier models.

Background you need

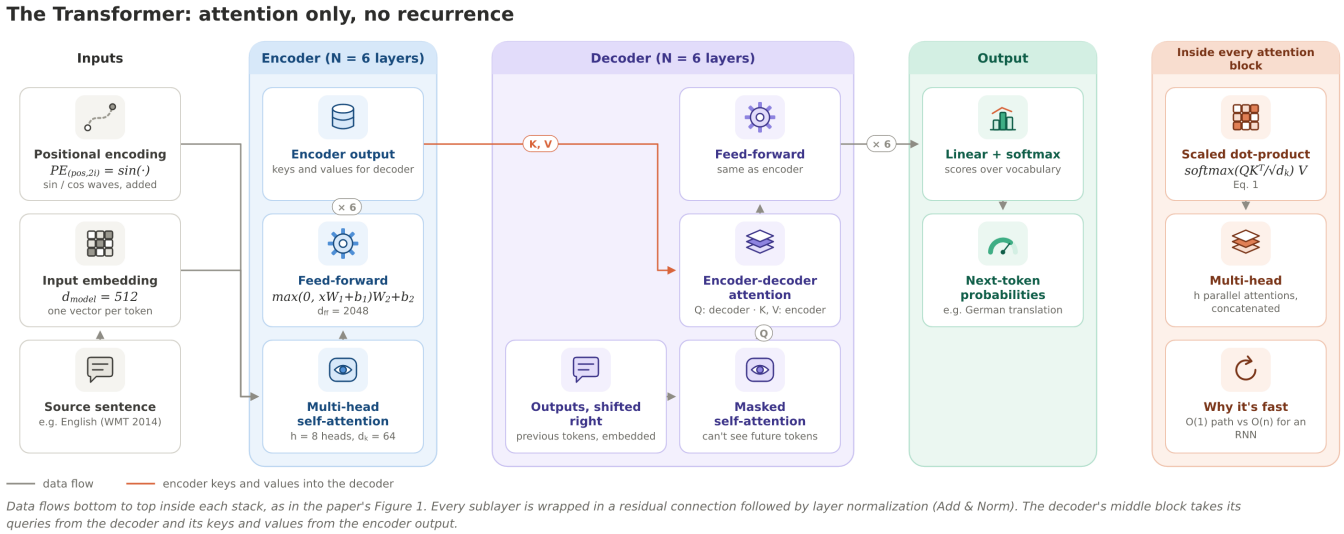
- **Sequence transduction**: turning one sequence into another, like a sentence in English into German.
- **RNN (recurrent network)**: reads tokens in order, carrying a memory forward, like reading with your finger on the page. It can't be parallelized across positions.
- **Attention**: a weighted look-up. Each word asks a question (**query**), every word offers a label (**key**) and content (**value**), and the answer is a weighted mix of values.
- **BLEU**: a 0-100 score of how close a translation is to human references. Higher is better.

The problem

Recurrent models process positions one after another, so training can't use the GPU's parallelism within a sentence, and a signal between the first and last word has to pass through every step in between. Earlier models used attention as an add-on to an RNN. The question here: **can attention alone do the whole job?**

What they did

- **Encoder-decoder built only from attention and feed-forward layers**: 6 encoder layers and 6 decoder layers, each sublayer wrapped in a residual connection and layer normalization.
- **Scaled dot-product attention**: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})\, V$. Dividing by $\sqrt{d_k}$ keeps the scores from growing too large.
- **Multi-head attention**: 8 heads attend in parallel, each with 64-dimensional keys and values, so the model can look for different kinds of relationships at once.
- **Positional encoding**: with no recurrence, word order is added back as sine and cosine waves of different frequencies.
- **Base model sizes**: $d_{model} = 512$, feed-forward $d_{ff} = 2048$, dropout 0.1, label smoothing 0.1.



Data flows bottom to top inside each stack, as in the paper's Figure 1. Every sublayer is wrapped in a residual connection followed by layer normalization (Add & Norm). The decoder's middle block takes its queries from the decoder and its keys and values from the encoder output.

Figure 1. The Transformer architecture, drawn from the paper's Figure 1 and Section 3. The decoder's encoder-decoder attention takes queries (Q) from the decoder and keys and values (K, V) from the encoder output.

Key findings

- **New state of the art, English→German**: the big Transformer scores 28.4 BLEU, more than 2.0 BLEU above the best previous models, including ensembles.
- **English→French**: 41.8 BLEU for a single big model, above every model and ensemble in Table 2.
- **Cheaper to train**: the base model costs $3.3 \cdot 10^{18}$ FLOPs, the lowest in Table 2, and still beats every earlier English→German result with 27.3 BLEU.
- **Fast**: the base model trains in about 12 hours and the big model in 3.5 days, on one machine with 8 NVIDIA P100 GPUs.
- **Why attention helps (Table 1)**: any two positions are connected in O(1) steps, versus O(n) for a recurrent layer, with O(1) sequential operations, so a whole sentence is processed in parallel.

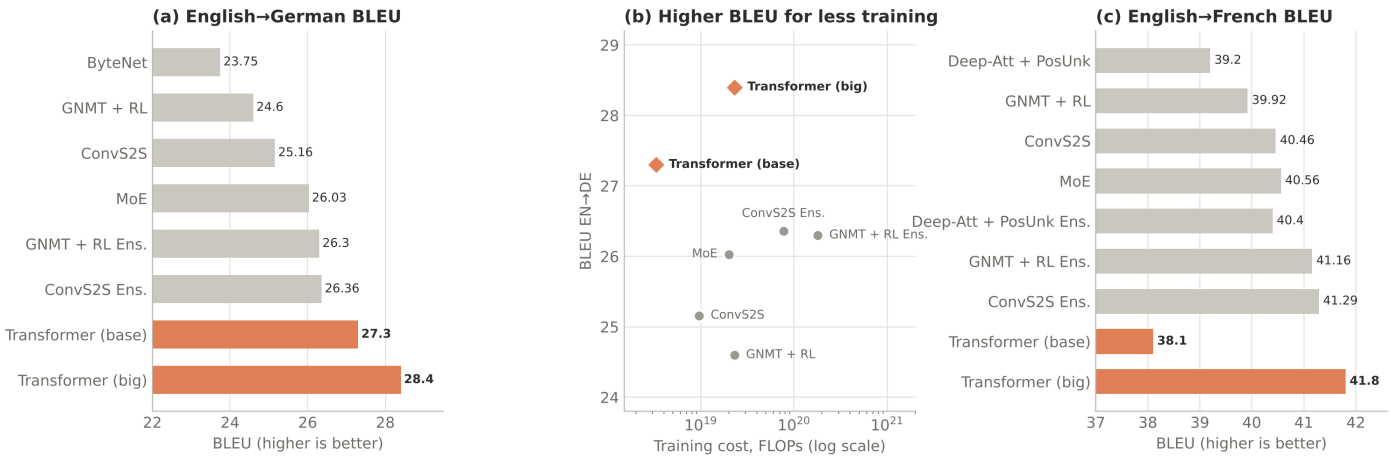


Figure 2. Table 2 of the paper on newstest2014. (a) English→German BLEU. (b) English→German BLEU against training cost. Both Transformers sit above every earlier model at equal or lower cost. (c) English→French BLEU.

Limitations

- Self-attention costs $O(n^2 \cdot d)$ per layer, which grows quickly for very long sequences. Section 4 suggests restricting attention to a local neighborhood, which the authors leave to future work.
- Evaluated on machine translation and English constituency parsing only.
- The base model is behind earlier ensembles on English→French (38.1 vs 41.29 BLEU). Only the big model leads there.

Why it matters

This paper introduced the architecture behind nearly every modern language model. Replacing step-by-step recurrence with **parallel attention** made training on huge datasets practical, which is the foundation of today's large language models and of vision-language-action models in robotics.